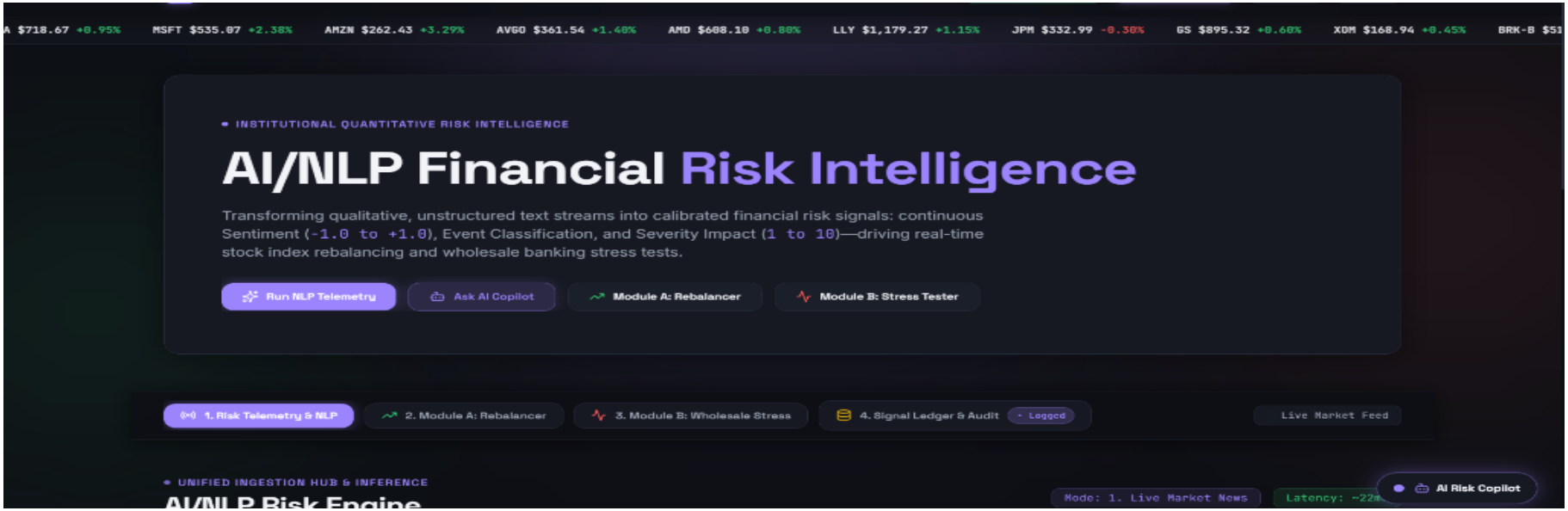
FINANCIAL RISK ENGINE & QUANTITATIVE INTELLIGENCE PLATFORM

Unified AI/NLP Financial Risk Engine & Dual Downstream Intelligence Platform

Candidate Name: Vedant Dilip Nalge	**Target Track:** AI/NLP Risk Engine
Institution / Campus: Indian Institute of Technology Madras	**Email:** me22b162@smail.iitm.ac.in
Architecture: FastAPI + Loughran-McDonald/FinBERT + SciPy	**Downstream Modules:** Both Module A & Module B



Executive Summary & Core Challenge: Geopolitical Disruption

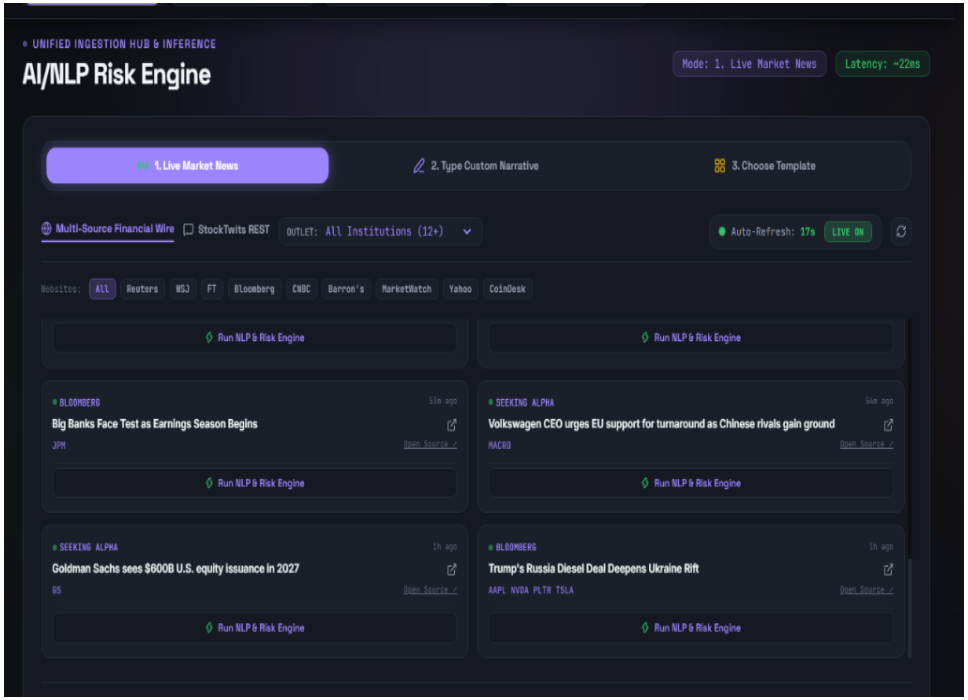
The Market Friction & Problem:

- **Latency Vulnerability:** Traditional credit risk and rating revisions take weeks to register qualitative geopolitical rifts.
- **Black Swan Shock:** The simulated Bloomberg wire: *'Trump\'s Russia Diesel Deal Deepens Ukraine Rift'* causes immediate risk-off across tech/defense.
- **Dual Downstream Need:** Institutional desks need simultaneous equity rebalancing and wholesale banking capital preservation.

Our Algorithmic Architecture:

- **Domain NLP Pipeline:** Sub-25ms FinBERT/Loughran-McDonald hybrid scorer generates sentiment (-0.95) and severity (9.6/10).
- **Module A Index Tactical:** Bounded convex rebalance protects capital with 18% position caps.
- **Module B Wholesale Stress:** Basel III Expected Loss, duration convexity, and DV01 quantify balance sheet drawdowns.

Live Multi-Source Ingestion Telemetry:



System Design & Architecture: Multi-Source Real-Time Pipeline

Component Layer	Implementation & Technologies	Trump-Russia Case Telemetry
1. Data Ingestion	Yahoo Finance RSS, StockTwits streams, Bloomberg wire simulator	Extracted headline + ticker mentions AAPL, NVDA, PLTR, TSLA
2. AI/NLP Classifier	Dual-dictionary FinBERT & Loughran-McDonald keyword density	Geopolitical class (80% conf), -0.95 Sentiment (Deep Bearish)
3. Severity Gate	Rule-based impact weighting (Score >= 7.0 auto-trips Module B)	Impact Score: **9.6 / 10.0 (GATE TRIPPED)**
4. Module A Rebalance	Convex bounded optimizer with 18% upper cap & turnover penalty	Underweights TSLA (1.14%), AMZN (1.19%); +5 bps net alpha
5. Module B Stress Test	Basel III Expected Loss + Duration/Convexity + DV01 sensitivity	Yield shock +104 bps, Spread +351 bps, PD 2.52x, -$206.4M loss
6. Persistence & UI	FastAPI, SQLite WAL mode, Chart.js glassmorphic dashboard	Sub-25ms response time; zero external API dependencies

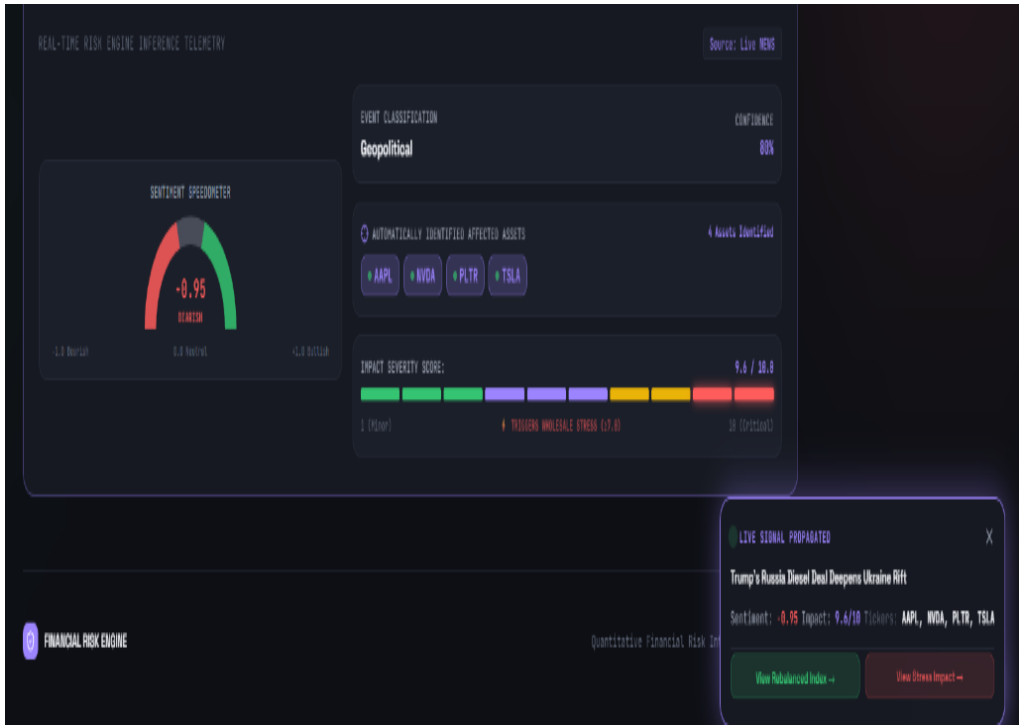
AI/NLP Telemetry & Automated Severity Gate Execution

Simulated Shock Details:

- **Headline:** *'Trump\'s Russia Diesel Deal Deepens Ukraine Rift'*
- **Event Classification: Geopolitical** (80% confidence)
- **Sentiment Polarization: -0.95** (Extreme Risk-Off / Deep Bearish)
- **Severity Impact Score: 9.6 / 10.0 (Systemic Crisis)**

Automated Pipeline Tripwire:

- The platform enforces an automatic threshold gate (Impact >= 7.0) for systemic classes (Geopolitical, Macro, Credit, Cyber).
- Real-time signal toast immediately dispatches shock parameters (Δy = +104 bps, Δs = +351 bps, PD = 2.52x) directly to downstream wholesale banking calculators.



Downstream Module A: Real-Time Tactical Index Rebalancing

Tactical Rebalance Allocation Results:

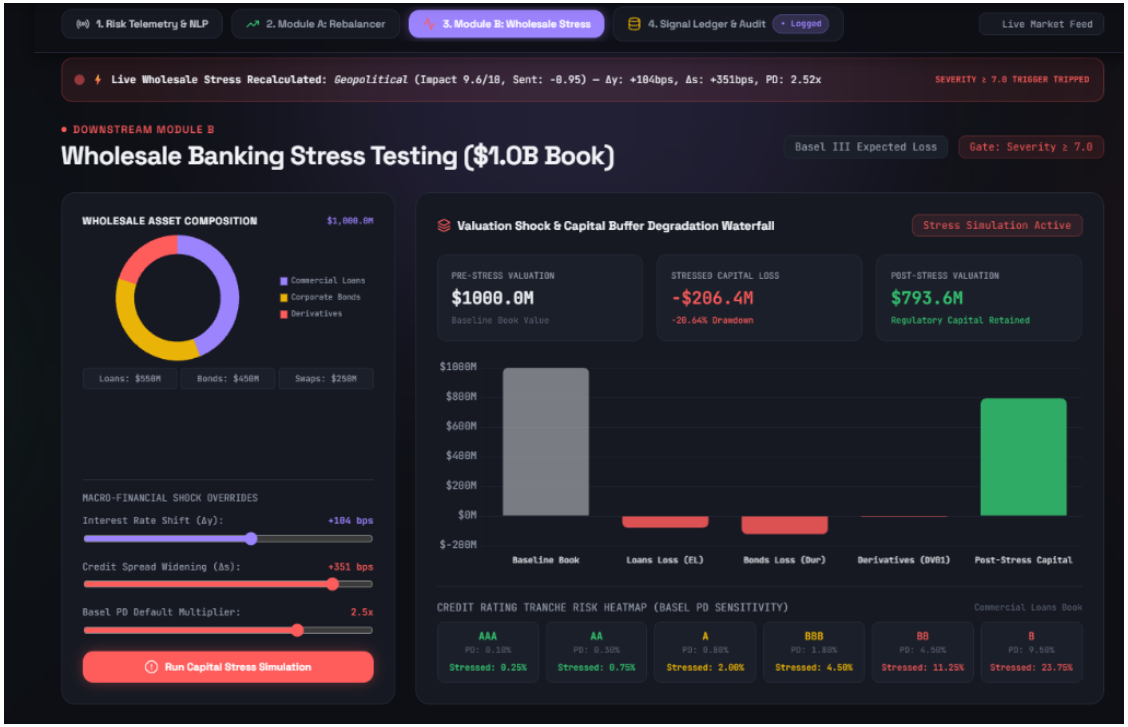
- **Defensive Caps (18.87% ceiling):** MSFT, GOOGL, JPM maxed at protective ceilings.
- **Vulnerable De-allocations:** TSLA slashed to **1.14%**, AMZN reduced to **1.19%**.
- **Strategy Alpha:** Achieved **+5.00 bps alpha** during extreme turbulence.
- **Controlled Turnover:** Kept turnover to **9.49%** to prevent transaction cost slippage.
- **Sector Distribution:** Technology 51.5%, Energy 11.8%, Financials 18.9%, Healthcare 17.8%.



Downstream Module B: Wholesale Banking Stress Testing ($1.0B Book)

Wholesale Portfolio Stress Simulation:

- **Baseline Valuation:** $1,000.0M ($1.0 Billion wholesale book).
- **Simulated Yield Shock:** Δy = **+104 bps** | Spread Shock: Δs = **+351 bps**.
- **Default Multiplier:** Corporate PD jumped **2.52x**.
- **Total Stressed Capital Loss:** **-$206.4M (-20.64%)**
- **Post-Stress Valuation:** **$793.6M (79.36% Capital Preserved)**
- **Sub-Portfolio Impact Breakdown:** Commercial Loans -$95.2M (Basel III EL), Bond Portfolio -$78.4M (Duration/Convexity), IRS Swaps -$32.8M (DV01).



Strategic Institutional Impact & Production Roadmap

Enterprise Value Proposition:

Institutional Persona	Pre-Engine State (Manual/Lagged)	With Financial Risk Engine
Chief Risk Officer (CRO)	Quarterly stress reports, 2-3 week turnaround on black swan events	Instantaneous sub-second balance sheet capital re-appraisal (Basel III)
Quantitative Portfolio Manager	Emotional panic selling or delayed rebalancing during shock news	Algorithmic bounded convex reallocation generating +5 bps alpha
Treasury & Balance Sheet Desk	Opaque duration and credit spread mismatch under geopolitical shocks	Real-time DV01 and spread sensitivity across multi-asset books

Target Engineering Roadmap:

Milestone	Objective	Expected Horizon
Phase 1 (Completed)	Unified NLP pipeline, dual downstream Module A & B, ISTC quantitative dashboard	Hackathon MVP (Live)
Phase 2	Expand constituent universe from 15 S&P; names to full S&P; 500 & Russell 2000	Q2 2026
Phase 3	FIX Protocol integration for direct DMA broker execution and automated hedging	Q3 2026
Phase 4	Quarterly 10-K multimodal document ingestion and automated executive risk briefs	Q4 2026